Ambient Polymorphism

Sam Estep

Abstract

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Introduction

PostgreSQL 11.22 (2023) returns zero for the following query (Simon and Lesage 1989):

NumAcc4.sql

```sql
SELECT stddev(
  CASE
    WHEN i = 0 THEN 10000000.2
    WHEN i % 2 = 1 THEN 10000000.1
    ELSE 10000000.3
  END :: double precision
)
FROM generate_series(0, 1000) AS i;
```

Why?

float.c

```c
Datum
float8_stddev_samp(PG_FUNCTION_ARGS)
{
    ArrayType  *transarray = PG_GETARG_ARRAYTYPE_P(0);
    float8     *transvalues;
    float8      N,
                sumX,
                sumX2,
                numerator;

    transvalues = check_float8_array(transarray, "float8_stddev_samp", 3);
    N = transvalues[0];
    sumX = transvalues[1];
    sumX2 = transvalues[2];

    /* Sample stddev is undefined when N is 0 or 1, so return NULL */
    if (N <= 1.0)
        PG_RETURN_NULL();

    numerator = N * sumX2 - sumX * sumX;
    CHECKFLOATVAL(numerator, isinf(sumX2) || isinf(sumX), true);

    /* Watch out for roundoff error producing a negative numerator */
    if (numerator <= 0.0)
        PG_RETURN_FLOAT8(0.0);

    PG_RETURN_FLOAT8(sqrt(numerator / (N * (N - 1.0))));
}
```

Bibliography

PostgreSQL Global Development Group. 2023. *PostgreSQL 11.22*. Released November 9. https://www.postgresql.org/docs/release/11.22/.

Simon, Stephen D., and James P. Lesage. 1989. “Assessing the Accuracy of ANOVA Calculations in Statistical Software.” *Computational Statistics & Data Analysis* 8 (3): 325–32. https://doi.org/10.1016/0167-9473(89)90048-0.